GenepoweRx

Genetic Science powered by K&H®

Patient Intake System

Technical Documentation

Version 1.0 | Beta

Prepared: March 2026

Stack: React.js · FastAPI · MongoDB · Python

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1. Project Overview

GenepoweRx is a clinical-grade patient intake system designed for genetic testing services, built under the brand "Genetic Science powered by K&H;®". The platform enables medical professionals and patients to initiate, manage, and track genetic testing workflows — from patient registration through test selection, consent capture, and administrative review.

■ **Mission:** Deliver a fast, HIPAA-conscious, mobile-friendly registration platform that connects patients with the appropriate genetic tests through a guided multi-step process, while providing administrators with real-time analytics and data management tools.

Project Name	GenepoweRx — Patient Intake System
Brand	Genetic Science powered by K&H;®
Status	Beta (Trial)
Target Users	Patients, Referring Doctors, Clinical Admins
Deployment	Local Development (localhost:3000 / localhost:8000)
Date Captured	March 24, 2026

2. Technology Stack

Layer	Technology	Purpose
Frontend	React.js (JSX)	UI components, routing, state management
Styling	CSS / App.css / index.css	Custom styling, responsive design
Backend	FastAPI (Python)	REST API, form handling, data validation
Database	MongoDB	Patient records, test data, submissions
Server	Uvicorn (ASGI)	FastAPI production/dev server
Package Mgr	npm	Frontend dependency management
Environment	Python .venv / pip	Backend dependency isolation
Language	JavaScript (ES6+) / Python 3.13	Full-stack development

3. Project Structure

The project follows a monorepo structure with a clear separation between the React frontend and the FastAPI backend. Below is the file tree as observed in the IDE:

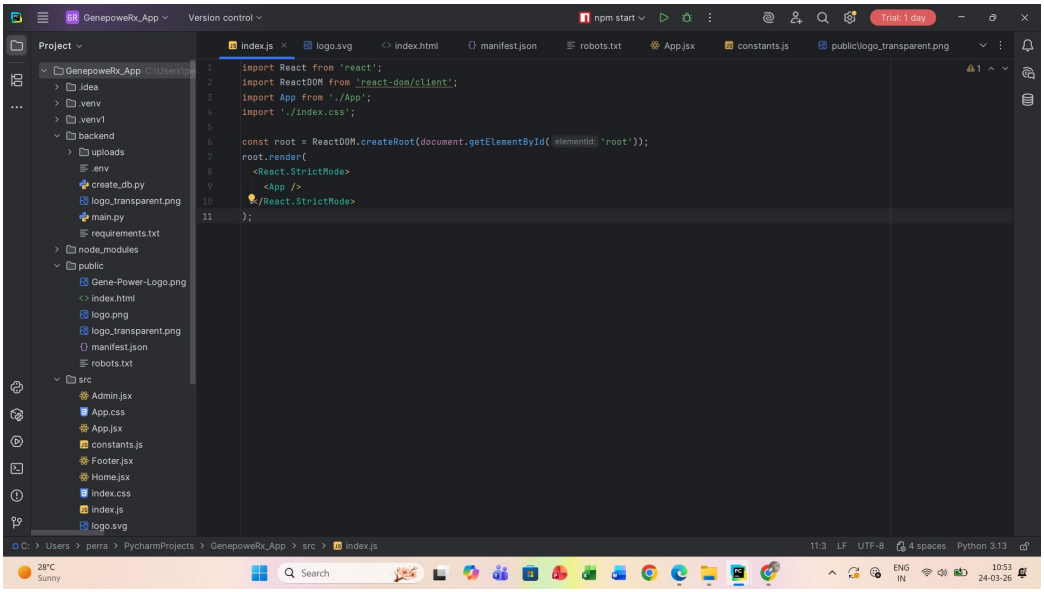

Figure 1a — Project structure (top): backend, public, and src folders

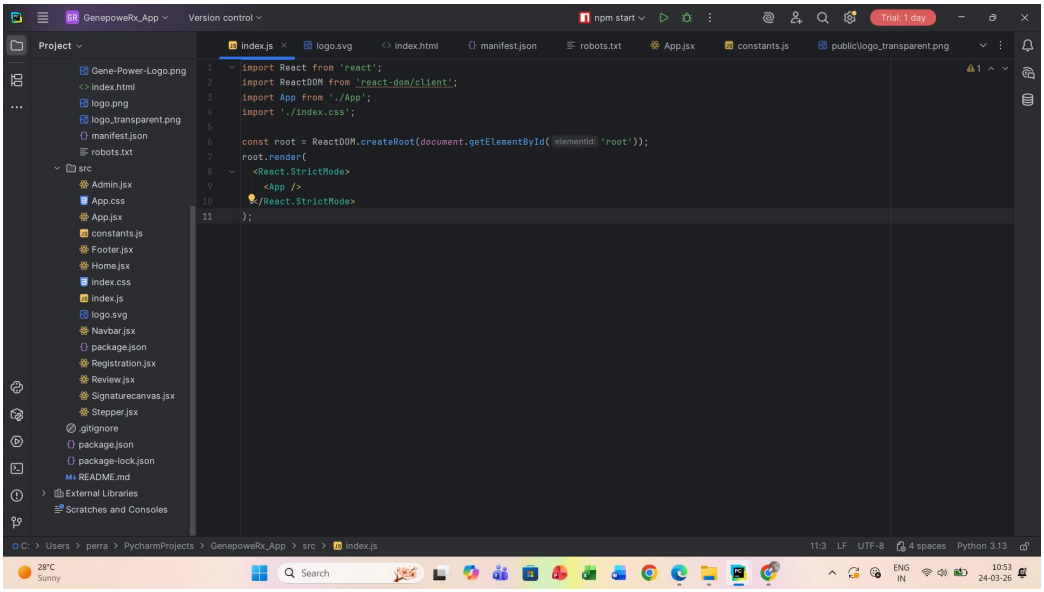

Figure 1b — Project structure (bottom): src component files and root config

Key Directories Explained:

Path / File	Description

`/backend/`	FastAPI application — main.py, create_db.py, requirements.txt, uploads/
`/public/`	Static assets served by React — index.html, logos, manifest.json, robots.txt
`/src/`	React source code — all JSX components and CSS files
`/src/components`	Admin.jsx, Home.jsx, Registration.jsx, Navbar.jsx, Footer.jsx, Stepper.jsx, Review.jsx, Signaturecanvas.jsx
`constants.js`	Shared constants — test names, GPX codes, configuration
`package.json`	npm dependency manifest for the React frontend
`requirements.txt`	Python pip dependencies for the FastAPI backend

4. How to Run the Application

The application consists of two independent servers that must both be running simultaneously — the FastAPI backend on port 8000 and the React frontend on port 3000. Open two separate terminal windows (e.g., PowerShell or Windows Terminal) and run the following commands:

4.1 Backend Server — FastAPI + Uvicorn

Navigate to the backend directory and start the FastAPI server with hot-reload enabled:

```
cd C:\Users\perra\PycharmProjects\GenepoweRx_App\backend

python -m uvicorn main:app --reload --port 8000
```

■ **Expected output:** Uvicorn will start and display *INFO: Application startup complete.* — the API is now live at **http://localhost:8000**. The **--reload** flag enables auto-restart on file changes.

4.2 Frontend Server — React (npm start)

Open a second terminal at the project root and launch the React development server:

```
cd C:\Users\perra\PycharmProjects\GenepoweRx_App

npm start
```

■ **Expected output:** The React app compiles and your browser automatically opens **http://localhost:3000**. Both servers must be running for full functionality (frontend ↔ backend communication).

Quick Start Summary

Step	Terminal	Command	URL
1	Terminal 1 (Backend)	cd backend python -m uvicorn main:app --reload --port 8000	localhost:8000
2	Terminal 2 (Frontend)	cd GenepoweRx_App npm start	localhost:3000

5. Application Pages & Features

5.1 Homepage

The homepage is the patient-facing landing page at **localhost:3000**. It presents the GenepoweRx brand with a full-width hero banner and clear call-to-action buttons.

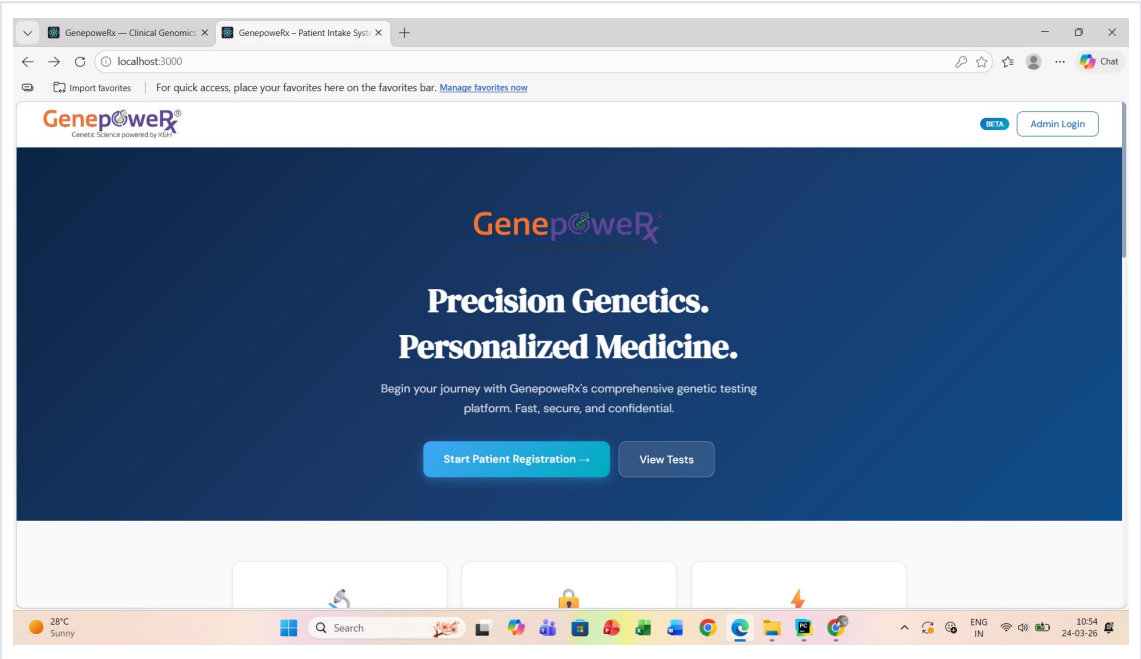

Figure 2 — Homepage Hero: "Precision Genetics. Personalized Medicine."

Hero Section Elements:

- GenepoweRx logo with "BETA" badge in the top-right
- Full-width dark navy gradient hero banner
- Main tagline: "Precision Genetics. Personalized Medicine."
- Sub-copy: "Begin your journey with GenepoweRx's comprehensive genetic testing platform."
- CTA Button 1 — "Start Patient Registration →" (navigates to /registration)
- CTA Button 2 — "View Tests" (anchor to test listing section)
- Admin Login button — top right for authorized staff

Feature Cards Section:

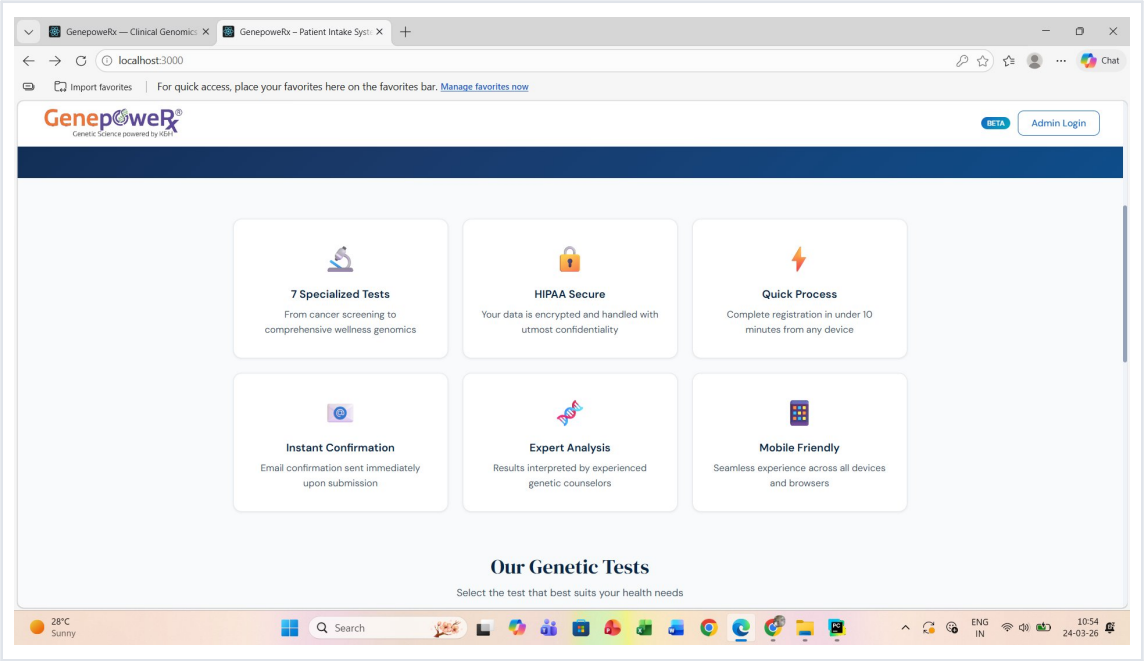

Figure 3 — Feature Cards: Key platform capabilities

Below the hero, six feature cards highlight the platform's core value propositions:

Feature	Description
■ 7 Specialized Tests	From cancer screening to comprehensive wellness genomics
■ HIPAA Secure	Data encrypted and handled with utmost confidentiality
■ Quick Process	Complete registration in under 10 minutes from any device
■ Instant Confirmation	Email confirmation sent immediately upon submission
■ Expert Analysis	Results interpreted by experienced genetic counselors
■ Mobile Friendly	Seamless experience across all devices and browsers

5.2 Patient Registration Form

Accessed via **localhost:3000/registration**, the registration form is a multi-step wizard that collects all necessary patient and clinical information before proceeding to consent and submission.

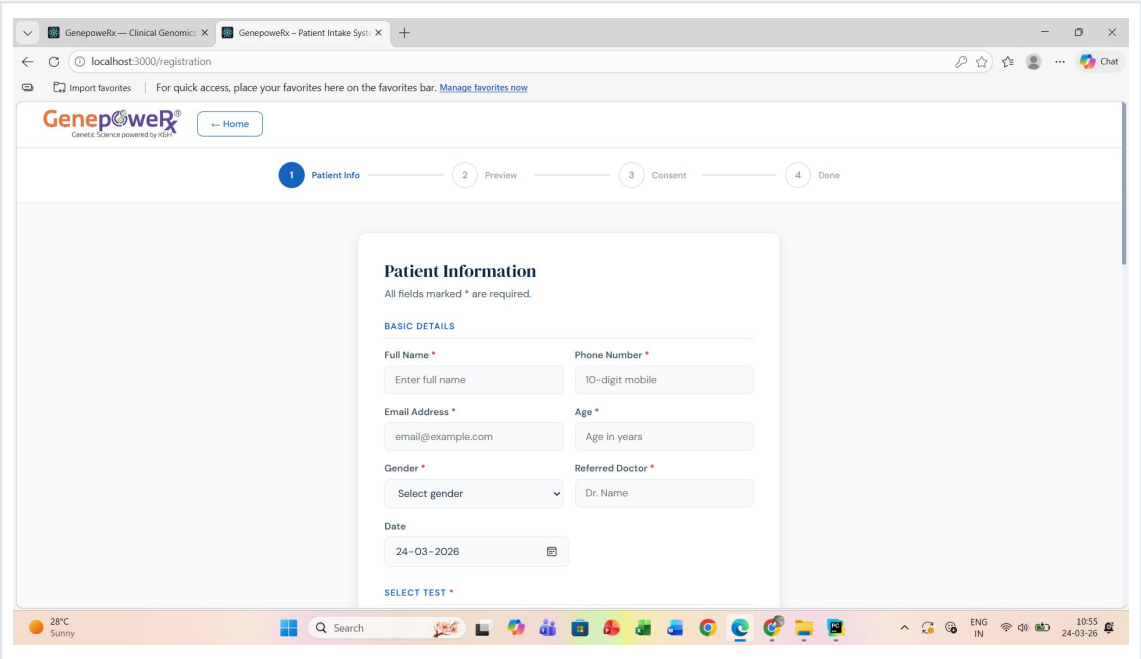

Figure 4 — Step 1: Patient Information Form

Multi-Step Workflow:

Step	Name	Description
1	Patient Info	Basic patient details + test selection + clinical notes
2	Preview	Review all entered information before proceeding
3	Consent	Digital signature capture and consent agreement
4	Done	Confirmation of successful submission

Form Fields — Patient Information (Step 1):

Field	Description	Input Type
Full Name *	Patient's complete legal name	Text input — required
Phone Number *	10-digit mobile number	Text input — required
Email Address *	Valid email for confirmation	Email input — required
Age *	Age in years	Numeric input — required
Gender *	Male / Female / Other	Dropdown selector — required
Referred Doctor *	Dr. Name of referring physician	Text input — required

Date	Auto-populated with today's date	Date picker — auto-filled
Select Test *	Genetic test selection (see Section 6)	Card-based selector — required
Reasons for Genetic Testing	Clinical justification / notes	Textarea — optional

5.3 Genetic Test Selection

Within the registration form, patients select from 7 available genetic tests displayed as interactive card tiles. Each card shows the test name and its unique GPX code.

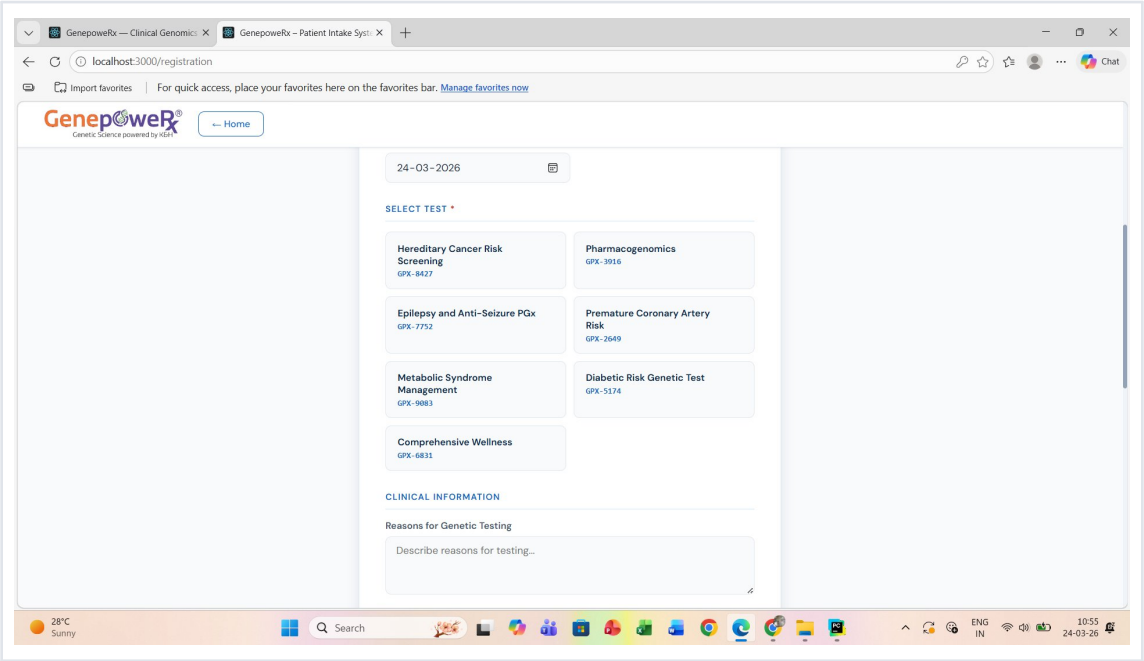

Figure 5 — Test Selection Cards with GPX codes

Available Tests:

Test Name	GPX Code	Description
Hereditary Cancer Risk Screening	**GPX-8427**	Cancer genetics panel for hereditary risk assessment
Pharmacogenomics	**GPX-3916**	Drug-gene interaction testing for medication optimization
Epilepsy and Anti-Seizure PGx	**GPX-7752**	Pharmacogenomic testing for anti-seizure medications
Premature Coronary Artery Risk	**GPX-2649**	Genetic risk assessment for early-onset heart disease
Metabolic Syndrome Management	**GPX-9083**	Genetic factors influencing metabolic health
Diabetic Risk Genetic Test	**GPX-5174**	Polygenic risk scoring for type 2 diabetes
Comprehensive Wellness	**GPX-6831**	Full genome wellness panel covering multiple health areas

5.4 Admin Dashboard — Overview

Accessible via **localhost:3000/admin**, the Admin Dashboard provides clinical staff with a real-time view of all patient submissions pulled directly from MongoDB. The dashboard has four main navigation sections: Overview, Patients, Analytics, and History.

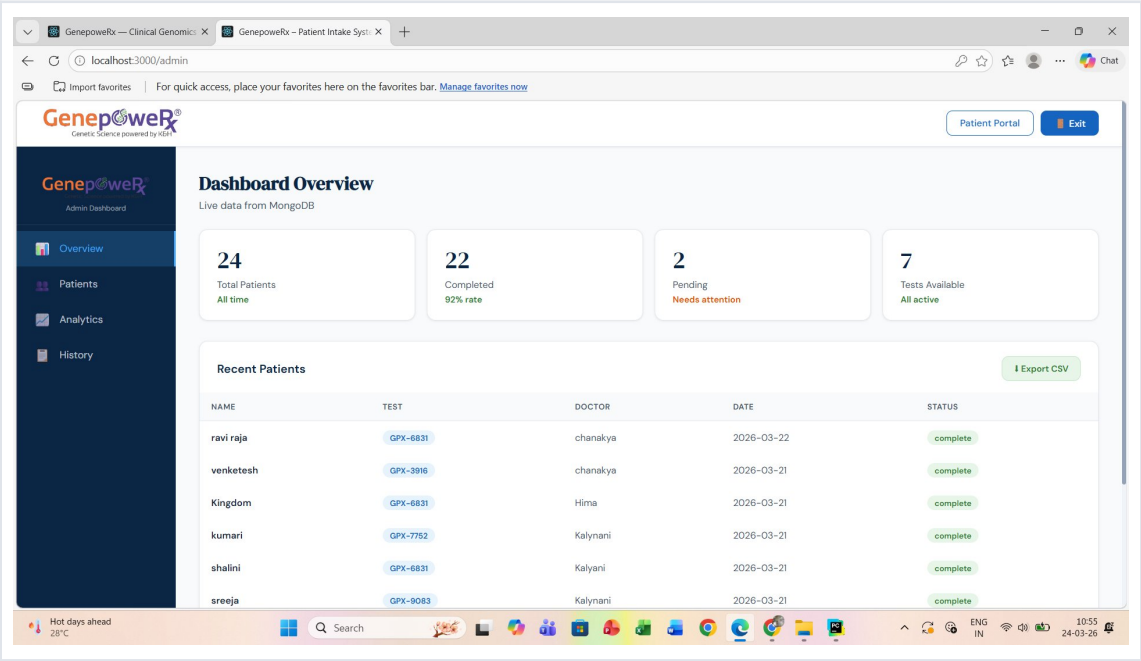

Figure 6 — Admin Dashboard Overview with live MongoDB data

Dashboard KPI Cards:

Metric	Value (Snapshot)	Label
Total Patients	**24**	All time registrations
Completed	**22**	92% completion rate
Pending	**2**	Needs attention
Tests Available	**7**	All active

Recent Patients Table:

The overview page shows the most recent patient submissions in a sortable table with columns: Name, Test (GPX code badge), Doctor, Date, and Status. Each record shows a green "complete" or amber "pending" status badge.

> ■ **Export CSV:** A one-click "Export CSV" button in the top-right of the Recent Patients table allows admins to download all patient data as a comma-separated file for external reporting or analysis.

5.5 Admin Dashboard — Analytics

The Analytics section (sidebar → Analytics) provides live data visualizations built from MongoDB aggregation queries, giving clinical managers insight into test distribution and demographics.

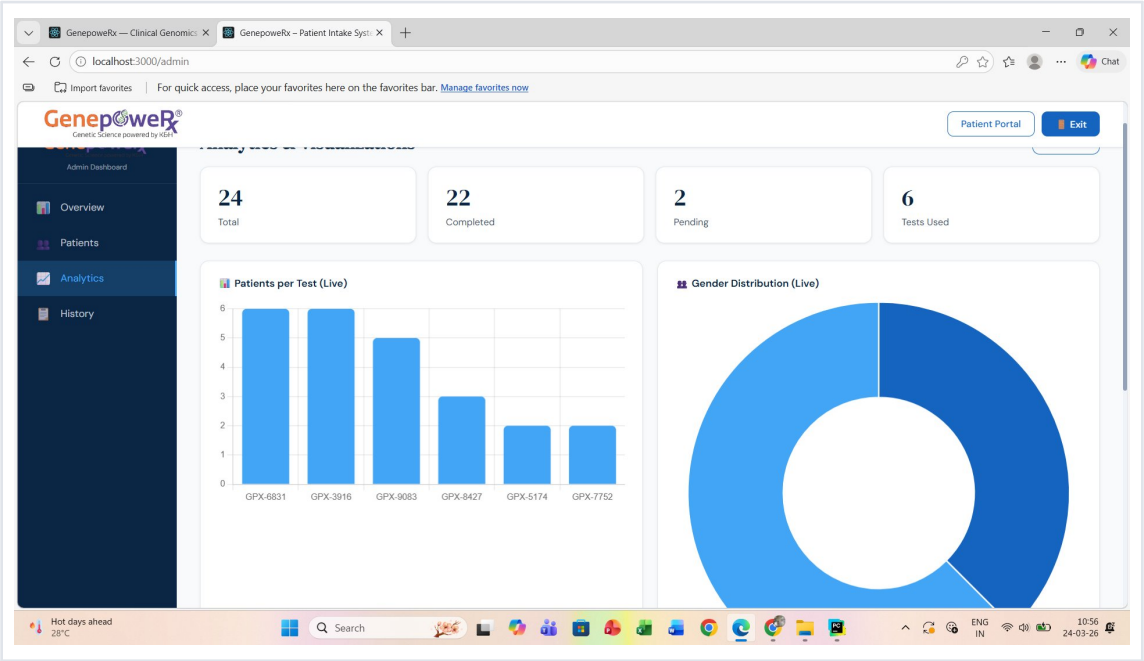

Figure 7 — Analytics: Bar chart (patients per test) + Gender Distribution pie chart

Analytics Charts Available:

- **Patients per Test (Live Bar Chart)** — Horizontal/vertical bar chart showing registration count per GPX test code. Highest volumes: GPX-6831, GPX-3916, GPX-9083.
- **Gender Distribution (Live Donut Chart)** — Pie/donut chart showing the male/female/other split across all registered patients.
- **Summary KPIs** — Total (24), Completed (22), Pending (2), Tests Used (6) displayed as card metrics at the top.

5.6 Admin Dashboard — Submission History

The History tab (sidebar → History) displays a complete chronological log of all patient submissions with sequential numbering (001, 002, ...) and full details.

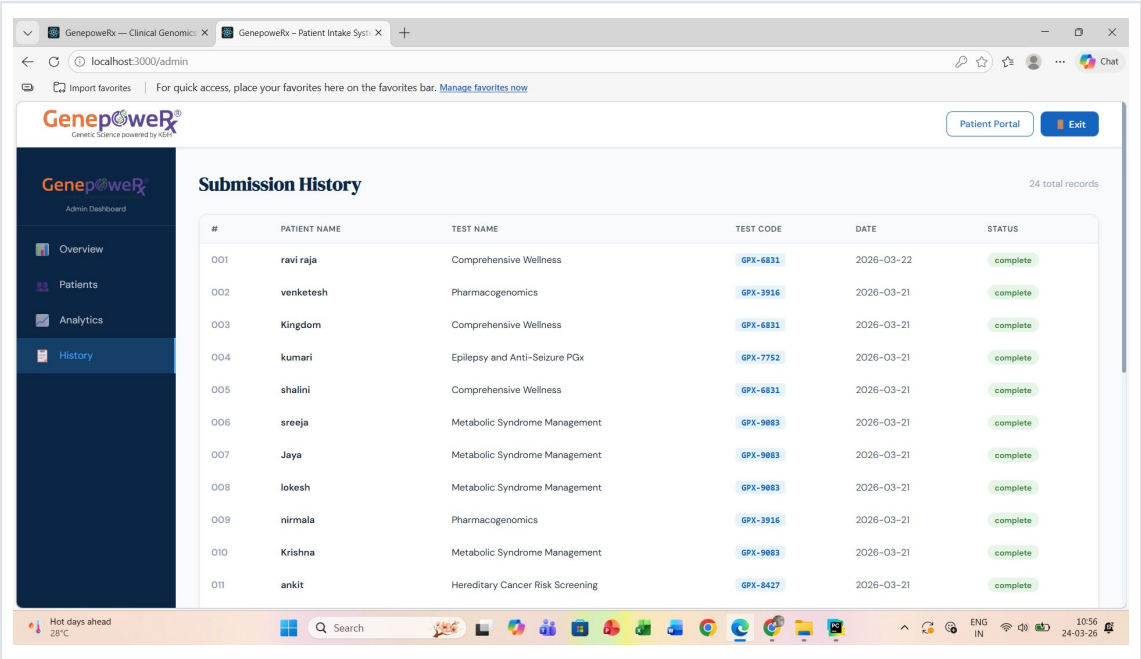

Figure 8 — Submission History: 24 total records with test names and status

History Table Columns:

Column	Description
#	Sequential record number (001, 002, ...)
Patient Name	Full name of the registered patient
Test Name	Full name of the selected genetic test
Test Code	GPX code displayed as a colored badge
Date	Submission date in YYYY-MM-DD format
Status	"complete" (green) or "pending" (amber) badge

6. Available Genetic Tests

GenepoweRx currently offers 7 specialized genetic tests. Each test has a unique GPX identifier used throughout the system for tracking, analytics, and reporting.

GPX-8427

Hereditary Cancer Risk Screening

A comprehensive panel assessing hereditary risk for multiple cancer types including breast, ovarian, colorectal, and other cancers. Identifies pathogenic variants in genes such as BRCA1/2, Lynch syndrome genes, and more.

GPX-3916

Pharmacogenomics

Drug-gene interaction testing that analyzes how a patient's genetic makeup affects their response to medications. Helps physicians optimize drug selection and dosing for safer, more effective treatment.

GPX-7752

Epilepsy and Anti-Seizure PGx

Specialized pharmacogenomic panel focused on anti-epileptic drugs. Identifies genetic variants that influence drug metabolism, efficacy, and adverse reaction risk for seizure management.

GPX-2649

Premature Coronary Artery Risk

Genetic risk assessment for early-onset coronary artery disease. Analyzes polygenic risk scores and monogenic variants associated with familial hypercholesterolemia and cardiomyopathy.

GPX-9083

Metabolic Syndrome Management

Examines genetic factors influencing obesity, insulin resistance, lipid metabolism, and hypertension to guide personalized lifestyle and pharmacological interventions.

GPX-5174

Diabetic Risk Genetic Test

Polygenic risk scoring for type 2 diabetes susceptibility combined with monogenic diabetes variant screening (MODY genes). Assists in early intervention and prevention strategies.

GPX-6831	Comprehensive Wellness
	A broad genomic wellness panel covering nutrition, fitness, sleep, mental health predispositions, and disease risk across multiple organ systems. The most comprehensive offering in the catalog.

7. Admin Dashboard Features

The Admin Dashboard (/admin) is a protected administrative interface for clinical staff. It provides four major capabilities through its sidebar navigation:

Overview

- Live KPI cards: Total Patients, Completed, Pending, Tests Available
- Recent Patients table with sortable columns
- One-click CSV export of all patient data
- Status badges: green (complete) / amber (pending)

Patients

- Full patient roster view
- Filter and search capabilities
- Individual patient record details

Analytics

- Live bar chart: Patients per Test (GPX codes)
- Live donut chart: Gender distribution
- Summary statistics cards
- Data sourced directly from MongoDB aggregations

History

- Complete chronological submission log
- 24 total records with sequential numbering
- Full test name, doctor, date, and status columns
- Search and filter by date range or status

■ **Access:** The Admin Dashboard is accessed via the "Admin Login" button on the homepage navbar. A separate "Exit" button (top-right of dashboard) and "Patient Portal" button allow quick navigation back to the patient-facing site.

8. API Architecture (Backend)

The backend is built with **FastAPI** (Python), served by **Uvicorn** on port 8000. FastAPI provides automatic OpenAPI documentation, async request handling, and Pydantic data validation. The API serves as the bridge between the React frontend and MongoDB.

Start Command:

```
python -m uvicorn main:app --reload --port 8000
```

Auto-Generated API Documentation:

- **Swagger UI** — `http://localhost:8000/docs` (Interactive API explorer)
- **ReDoc** — `http://localhost:8000/redoc` (Alternative API documentation)
- **OpenAPI JSON** — `http://localhost:8000/openapi.json` (Machine-readable schema)

Backend Files:

File	Purpose
`main.py`	FastAPI app entry point — defines all API routes and CORS config
`create_db.py`	MongoDB connection setup and database initialization script
`requirements.txt`	Python dependencies: fastapi, uvicorn, pymongo, python-dotenv, etc.
`.env`	Environment variables — MongoDB connection URI, secret keys (not committed)
`uploads/`	Directory for any file uploads processed by the backend

9. Database (MongoDB)

GenepoweRx uses **MongoDB** as its primary data store. The Admin Dashboard explicitly confirms "Live data from MongoDB" on the overview page. MongoDB's document model is well-suited for storing flexible patient records with varying clinical fields.

Typical Patient Document Structure:

```
{
"_id": ObjectId("..."),
"name": "ravi raja",
"phone": "9XXXXXXXXX",
"email": "patient@example.com",
"age": 35,
"gender": "Male",
"referredDoctor": "chanakya",
"date": "2026-03-22",
"testCode": "GPX-6831",
"testName": "Comprehensive Wellness",
"reasons": "Family history of metabolic disorders",
"status": "complete",
"consentSigned": true,
"submittedAt": ISODate("2026-03-22T...")
}
```

■ **Live Analytics:** The Analytics section uses MongoDB aggregation pipelines to compute counts per test code (for the bar chart) and gender distribution (for the donut chart) in real-time without caching.

10. Security & Compliance

GenepoweRx is designed with clinical data standards in mind. The following security considerations are reflected in the platform:

HIPAA-Conscious Design

Patient data is described as "encrypted and handled with utmost confidentiality." The platform is positioned for HIPAA-compliant deployment.

Digital Consent Capture

Step 3 of the registration workflow (Signaturecanvas.jsx) captures a digital signature as proof of informed consent before submission.

Environment Variables

Sensitive credentials (MongoDB URI, keys) are stored in .env files excluded from version control via .gitignore.

Admin Access Control

The Admin Dashboard is separated from the patient portal and requires authentication ("Admin Login" button with access control).

CORS Configuration

FastAPI backend includes CORS middleware to restrict API access to authorized frontend origins only.

Data Validation

FastAPI's Pydantic models enforce strict data type validation on all incoming patient form submissions.

No PII in URLs

Patient information is submitted via POST requests, never exposed in URL parameters.

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